Solution

(i) $$\int_0^{\frac{\pi}{2}} \sin^4 x \cos^6 x \, dx = \frac{(6-1)}{(6+4)} \cdot \frac{(6-3)}{(6+4-2)} \cdot \frac{(6-5)}{(6+4-4)} \cdot \frac{(4-1)}{(4)} \cdot \frac{(4-3)}{(4-2)} \cdot \frac{\pi}{2}$$

$$= \frac{(5)}{(10)} \frac{(3)}{(8)} \frac{(1)}{(6)} \frac{(3)}{(4)} \frac{(1)}{(2)} \frac{\pi}{2} = \frac{3\pi}{512}$$

Also, $$\int_0^{\frac{\pi}{2}} \sin^4 x \cos^6 x \, dx = \int_0^{\frac{\pi}{2}} \sin^6 x \cos^4 x \, dx = \frac{(3)}{(10)} \frac{(1)}{(8)} \frac{(5)}{(6)} \frac{(3)}{(4)} \frac{(1)}{(2)} \frac{\pi}{2} = \frac{3\pi}{512}$$

(ii) $$\int_0^{\frac{\pi}{2}} \sin^5 x \cos^4 x \, dx = \frac{(3)}{(9)} \frac{(1)}{(7)} \frac{(4)}{(5)} \frac{(2)}{(3)} = \frac{(4)}{(9)} \frac{(2)}{(7)} \frac{(1)}{(5)} = \frac{8}{315}$$

Also, $$\int_0^{\frac{\pi}{2}} \sin^5 x \cos^4 x \, dx = \int_0^{\frac{\pi}{2}} \sin^4 x \cos^5 x \, dx = \frac{(4)}{(9)} \frac{(2)}{(7)} \frac{(1)}{(5)} = \frac{8}{315}$$

Example 9.40

Evaluate $\int_0^{2a} x^2 \sqrt{2ax - x^2} \, dx$.

Solution

Put $x = 2a\cos^2\theta$. Then, $dx = -4a\cos\theta\sin\theta d\theta$.

When $x = 0$, $2a\cos^2\theta = 0$ and so $\theta = \frac{\pi}{2}$. When $x = 2a$, $2a\cos^2\theta = 2a$ and so $\theta = 0$.

Hence, we get

$$I = \int_0^{2a} x^2 \sqrt{2ax - x^2} \, dx$$

$$= \int_{\frac{\pi}{2}}^{0} 4a^2 \cos^4\theta \sqrt{4a^2\cos^2\theta - 4a^2\cos^4\theta}(-4a\cos\theta\sin\theta)d\theta$$

$$= \int_0^{\frac{\pi}{2}} 4a^2\cos^4\theta 2a\cos\theta\sin\theta(4a\cos\theta\sin\theta)d\theta$$

$$= 32a^4 \int_0^{\frac{\pi}{2}} \cos^6\theta\sin^2\theta \, d\theta$$

$$= 32a^4 \times \frac{1}{8} \times \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = \frac{5\pi a^4}{8}.$$

Example 9.41

Evaluate $\int_0^1 x^5 \left(1 - x^2\right)^5 dx$.

Solution

Put $x = \sin\theta$. Then, $dx = \cos\theta d\theta$.

When $x = 0$, $\sin\theta = 0$ and so $\theta = 0$. When $x = 1$, $\sin\theta = 1$ and so $\theta = \frac{\pi}{2}$.

Hence, we get

$$I = \int_0^{\frac{\pi}{2}} \sin^5\theta \left(1 - \sin^2\theta\right)^5 \cos\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \sin^5\theta\cos^{11}\theta \, d\theta = \frac{10}{16} \times \frac{8}{14} \times \frac{6}{12} \times \frac{4}{10} \times \frac{2}{8} \times \frac{1}{6} = \frac{1}{336}.$$

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